

INTRODUCTION TO IP ADDRESSING THEORY

1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.



Difference Between IPv4 and IPv6

IPv4	IPv6
Uses a 32-bit address.	Uses a 128-bit address.
Address format is 192.168.1.1.	Address format is 2001:0db8:85a3::8a2e:0370:7334.
Supports about 4.3 billion unique IP addresses.	Supports 340 undecillion unique IP addresses, enough for future devices.
Uses decimal numbers separated by dots.	Uses hexadecimal numbers separated by colons.
Security (IPSec) is optional.	IPSec support is built into the protocol.
Requires NAT because IPv4 addresses are limited.	NAT is generally not required because of the large address space.

2. Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.



Static IPv4 Address Assignment (192.168.10.0/24 Network)

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.



4. You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.



5. Use an online subnet calculator tool (such as <https://www.subnet-calculator.com/>) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.



Insert the Chrome browser screenshot of <https://www.subnet-calculator.com> showing the calculation for 192.168.1.0/27. The page should display:

- Subnet mask: 255.255.255.224
- Total subnets
- Usable hosts per subnet: 30
- Network address: 192.168.1.0
- Broadcast address: 192.168.1.31
- First few subnet ranges: 192.168.1.1–192.168.1.30, 192.168.1.32–192.168.1.62 ...)

Code ^

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The manual subnet calculation matched the output of the online subnet calculator.